

CamoFarmer: A Lightweight Computer Vision Model for Camouflaged Crop Detection

Proponents & Adviser

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Project Structure

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CamoFarmer/
├── knowledge_distillation/      # Knowledge distillation implementations
│   ├── ssd/                   # SSD-based knowledge distillation
│   │   ├── main_kd.py
│   │   └── solver_kd.py
│   ├── yolov8/                # YOLOv8-based knowledge distillation
│   │   ├── distill_model.py
│   │   ├── distill_train.py
│   │   └── main.py
│   └── rtdetr/                # RT-DETR-based knowledge distillation
│       ├── distill_model.py
│       ├── distill_train.py
│       └── rtdetr_main.py
├── pruning/                   # Model pruning techniques
│   ├── main_prune.py          # Pruning main script (SSD)
│   ├── solver_prune.py        # Pruning solver (SSD)
│   ├── prune_helpers.py       # Pruning utility functions
│   ├── prune_validate.py      # Validation tools for ultralytics-based model validation
│   └── ultralytics_pruner.py  # YOLOv8 and RT-DETR pruning
├── backbone_replacement/      # Config files of different backbones for each model
│   ├── ssd/
│   ├── rtdetr/
│   └── yolov8/
├── CamoFarmer User Manual.pdf  # User manual
└── CamoFarmer Technical Manual.pdf # Technical manual

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Overview

Camouflaged object detection (COD) has emerged as a key innovation in agriculture, addressing the challenge of identifying crops that blend into their natural environments.

This study introduces **CamoFarmer**, a lightweight detection model optimized for camouflaged crop detection in resource-constrained settings.

We adapted three general object detection models — **SSD**, **YOLOv8**, and **RT-DETR** — using:

- **Lightweight backbone replacement**
- **Network pruning**
- **Knowledge distillation**

These techniques aimed to reduce model complexity while maintaining strong detection performance.

Key Results

Experiments were conducted on **TomatOD**, **CCROP**, and **CamoCrops** datasets.
After extensive evaluation, **RT-DETR** was chosen as the base model for **CamoFarmer**.

The final optimized model:

- **Pruned by 30% across all layers**
- **Parameter reduction:** 22.81% (from 32.8M → 25.35M)
- **FLOPs:** Unchanged at 29.57G
- **Performance:**
 - Up to **+5% mAP** improvement on some datasets
 - Minimal accuracy decreases on others

Complete Experiment Results

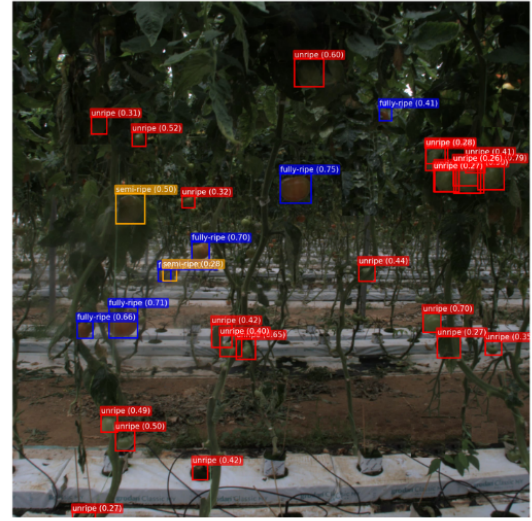
Model & Strategy	TomatOD mAP50	Params (M)	FLOPs (G)	CCROP mAP50	Params (M)	FLOPs (G)	CamoCrops mAP50	Params (M)	FLOPs (G)
SSD -- Baseline	67.70	24.01	61.11	87.09	23.88	60.96	56.77	25.62	62.30
YOLOv8n -- Baseline	71.92	3.01	8.10	94.78	3.01	8.10	52.21	3.01	8.10
YOLOv8l -- Baseline	82.37	43.61	164.80	94.49	43.61	164.80	57.55	43.61	164.80
RT-DETR -- Baseline	57.59	32.81	29.58	94.87	32.81	29.57	55.99	32.84	29.59
SSD -- Shufflenet V2	25.10	1.98	1.07	73.31	1.89	1.00	46.15	3.08	1.85
SSD -- MobileNet V2	24.80	2.82	1.31	72.17	2.75	1.29	49.21	3.72	1.59
SSD -- EfficientNet	20.80	3.88	1.54	71.45	3.84	1.51	45.49	4.30	1.76
YOLOv8n -- Shufflenet V2	70.82	3.30	8.00	92.94	3.30	8.00	46.68	3.31	8.00
YOLOv8n -- MobileNet V2	74.81	4.35	10.90	93.70	4.35	10.80	45.94	4.35	10.90
YOLOv8n -- EfficientNet	76.69	6.53	12.90	94.70	6.53	13.00	43.11	6.54	13.00
YOLOv8l -- Shufflenet V2	75.27	25.41	81.80	93.09	25.41	81.80	49.16	25.43	82.10
YOLOv8l -- MobileNet V2	77.05	26.55	84.70	93.69	26.55	84.70	52.08	26.57	84.90
YOLOv8l -- EfficientNet	79.17	29.39	87.50	94.53	29.39	87.50	47.19	29.41	87.80
RT-DETR -- Shufflenet V2	19.19	21.29	19.44	89.08	21.29	19.44	46.90	21.31	19.45
RT-DETR -- MobileNet V2	19.80	21.49	19.76	90.63	21.49	19.76	44.86	21.51	19.78
RT-DETR -- EfficientNet	24.76	26.03	20.71	91.08	26.03	20.71	46.27	26.05	20.73

Model & Strategy	TomatoOD mAP50	Params (M)	FLOPs (G)	CCROP mAP50	Params (M)	FLOPs (G)	CamoCrops mAP50	Params (M)	FLOPs (G)
SSD -- Pruning (All) -- 0.03	69.40	23.29	61.10	86.52	24.85	60.96	55.40	23.16	62.30
SSD -- Pruning (All) -- 0.05	68.40	22.81	61.10	86.85	24.34	60.96	54.95	22.69	62.30
SSD -- Pruning (All) -- 0.1	69.00	21.61	61.10	86.83	23.06	60.96	55.35	21.49	62.30
SSD -- Pruning (All) -- 0.3	68.90	16.81	61.10	86.41	17.93	60.96	54.38	16.72	62.30
SSD -- Pruning (All) -- 0.5	69.90	12.01	61.10	86.19	12.81	60.96	54.76	11.94	62.30
YOLOv8n -- Pruning (All) -- 0.03	75.16	2.92	8.10	94.53	2.92	8.10	46.79	2.92	8.10
YOLOv8n -- Pruning (All) -- 0.05	74.11	2.86	8.10	94.64	2.86	8.10	46.73	2.86	8.10
YOLOv8n -- Pruning (All) -- 0.1	10.79	2.71	8.10	60.83	2.71	8.10	36.35	2.71	8.10
YOLOv8n -- Pruning (All) -- 0.3	0.15	2.11	8.10	0.00	2.11	8.10	1.58	2.11	8.10
YOLOv8n -- Pruning (All) -- 0.5	0.07	1.51	8.10	0.01	1.51	8.10	0.01	1.51	8.10
YOLOv8l -- Pruning (All) -- 0.03	82.38	42.32	164.80	94.79	42.32	164.80	47.67	42.33	164.80
YOLOv8l -- Pruning (All) -- 0.05	82.45	41.45	164.80	94.58	41.45	164.80	48.88	41.46	164.80
YOLOv8l -- Pruning (All) -- 0.1	13.96	39.27	164.80	76.68	39.27	164.80	40.52	39.28	164.80
YOLOv8l -- Pruning (All) -- 0.3	0.03	30.55	164.80	0.44	30.55	164.80	2.68	30.56	164.80
YOLOv8l -- Pruning (All) -- 0.5	0.16	21.83	164.80	0.01	21.83	164.80	0.03	21.84	164.80
RT-DETR -- Pruning (All) -- 0.03	64.63	32.06	29.58	94.94	32.06	29.57	52.44	32.09	29.59
RT-DETR -- Pruning (All) -- 0.05	61.13	31.56	29.58	94.92	31.56	29.57	52.97	31.59	29.59
RT-DETR -- Pruning (All) -- 0.1	64.89	30.32	29.58	94.88	30.32	29.57	53.17	30.35	29.59
RT-DETR -- Pruning (All) -- 0.3	60.96	25.35	29.58	95.02	25.35	29.57	53.84	25.38	29.59
RT-DETR -- Pruning (All) -- 0.5	58.40	20.38	29.58	95.18	20.38	29.57	52.72	20.41	29.59

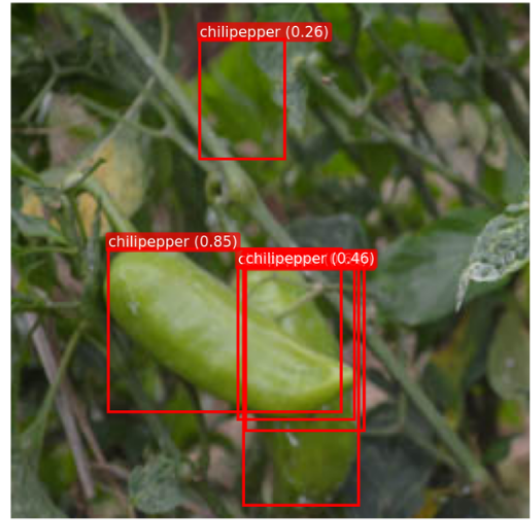
Model & Strategy	TomatoOD mAP50	Params (M)	FLOPs (G)	CCROP mAP50	Params (M)	FLOPs (G)	CamoCrops mAP50	Params (M)	FLOPs (G)
SSD -- SSD-EfficientNet	25.80	3.88	1.54	73.40	3.84	1.52	44.27	4.30	1.76
SSD -- SSD-MobileNet V2	23.50	2.82	1.31	73.77	2.75	1.29	48.19	3.72	1.59
SSD -- SSD-ShuffleNet V2	20.40	1.98	1.07	70.01	1.89	1.00	44.60	3.08	1.85
YOLOv8n -- YOLOv8l - YOLOv8n(ShuffleNet V2)	72.87	3.30	8.00	92.01	3.30	8.00	44.76	3.31	8.00
YOLOv8n -- YOLOv8l- YOLOv8n(EfficientNet)	79.24	6.53	12.90	94.82	6.53	13.00	42.34	6.54	13.00
YOLOv8n -- YOLOv8l- YOLOv8n(MobileNet V2)	75.65	4.35	10.90	93.52	4.35	10.80	43.96	4.35	10.90
YOLOv8l -- YOLOv8l - YOLOv8l (ShuffleNet V2)	74.78	25.41	81.80	92.83	25.41	81.80	43.15	25.43	82.10
YOLOv8l -- YOLOv8l- YOLOv8l(EfficientNet)	78.34	29.39	87.50	94.16	29.39	87.50	40.73	29.41	87.80
YOLOv8l -- YOLOv8l- YOLOv8l(MobileNet V2)	76.10	26.55	84.70	93.55	26.55	84.70	44.84	26.57	84.90
RT-DETR -- RTDETR-EfficientNet	27.32	26.03	20.71	97.98	26.03	20.71	43.40	26.05	20.73
RT-DETR -- RTDETR-MobileNet V2	20.71	21.49	19.76	97.47	21.49	19.76	45.32	21.51	19.78
RT-DETR -- RTDETR-ShuffleNet V2	22.92	21.29	19.44	96.74	21.29	19.44	44.50	21.31	19.45

Qualitative Results

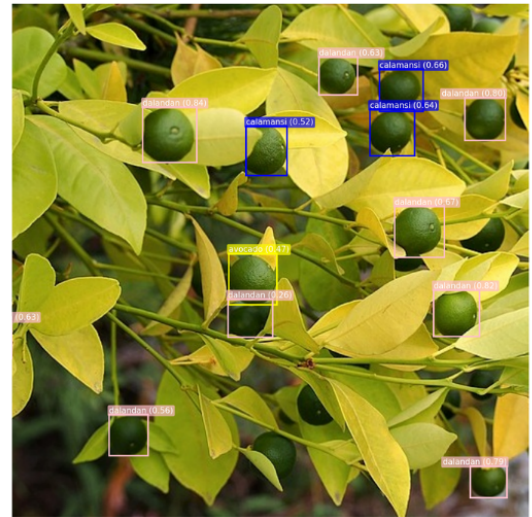
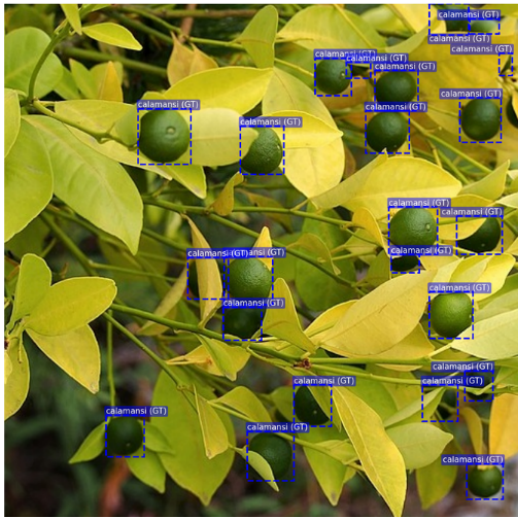
TomatoD



CCROP



CamoCrops



(a) Ground Truth

(i) RT-DETR 30%
All Conv.

Figure: shows qualitative differences after pruning on RT-DETR-I with 30% all-convolutional layer pruning. On TomatoD, duplicate detections and occasional misclassifications increase in dense foliage. On CCROP, we observe extra boxes and tighter clusters around single objects, indicating reduced localization precision. On CamoCrops, the main failure mode is

missed small or partially occluded crops. Overall, pruning trades computation for accuracy. The effect is strongest where camouflage and occlusion are severe, and the dataset is more complex.

Trained Weights

Trained model weights are available here:

[Download from Google Drive](#)